Mardhya Malik Nurbani

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SUMMARY

Ex-banking professional turned AI Engineer with a strong foundation in data science, gained through the Hacktiv8 Data Science Bootcamp and hands-on project experience. Over two years of experience in AI, currently building and deploying production LLM systems at PLN Icon Plus — agentic chatbots, multi-tenant RAG, generative content pipelines, and voice assistants — on self-hosted GPU infrastructure (NVIDIA H200, vLLM), as well as a computer vision safety-monitoring platform with human-in-the-loop retraining. Background in finance brings a unique strength in client-focused problem solving and domain-specific understanding of financial data.

EDUCATION

Hacktiv8 Bootcamp — **Remote**
Data Science Program. (Transcript) — *11/2023 – 02/2024*

Digitalent Kominfo — **Remote**
Intro to Full Stack Developer — *05/2021 – 07/2021*

Telkom University — **Bandung, Indonesia**
Bachelor of Telecommunication Engineering — *2015 – 2020*

WORK EXPERIENCE

PLN Icon Plus — **Jakarta, Indonesia**
AI Engineer — *April 2025 – Present*

- Enterprise Generative AI Platform — LLM, RAG & Agentic Solutions
  Built and deployed a family of LLM-powered services for PLN business units on a shared stack: self-hosted LLMs, embedding, and reranker models on NVIDIA H200 via vLLM behind a LiteLLM gateway, LangGraph/LangChain orchestration, FastAPI backends, Milvus hybrid search, Langfuse observability, PostgreSQL, MinIO, and Docker.
  - Conversational Agent — PLN Mobile Customer Service Chatbot
    - Designed a LangGraph multi-agent workflow with intent classification, conditional routing, and a ReAct agent delegating to specialized sub-agents for billing and account data, network outage information, and complaint tracking with ticket generation, plus WebSocket-based live agent handoff.
    - Built a multi-stage RAG pipeline (query enhancement from history, relevance grading, completeness scoring) on Milvus with a self-hosted embedding model and a fallback chain: vector retrieval → query reformulation → web search.
    - Implemented short-term (auto-trimmed history) and long-term memory (structured conversation summaries in PostgreSQL) with PostgresSaver checkpointing.
    - Benchmarked TTFT and total latency across concurrency levels and prompt tiers on H200 to select the inference stack; traced every LLM call, tool call, and graph node in Langfuse for latency and conversation-level analytics (intent distribution, resolution rate).
  - Document Intelligence — Elarch Public (Multi-Tenant RAG over Archives)
    - Built a multi-tenant RAG service for PLN's electronic archive, with an 8-node LangGraph StateGraph and a hybrid rule/LLM router serving four modes: Q&A, summary, multi-document comparison, and document discovery.
    - Implemented hybrid retrieval in Milvus (dense HNSW + BM25 sparse, fused with RRF) with a self-hosted reranker, and per-tenant isolation via partition keys; introduced table-aware chunking that significantly improved retrieval accuracy on tabular content.
    - Built an asynchronous ingestion pipeline (MinIO → Redis queue → vision-model OCR via Docling → chunking, LLM summarization, embedding → Milvus).
    - Delivered SSE streaming with live inline citations.

  - Generative AI for Learning & Development — PLN Pusdiklat
    - Built a Clean Architecture (ports and adapters) FastAPI backend with LangGraph/LangChain agents on Gemini 2.5 Pro (with optional bring-your-own-key OpenAI per request) for competency

assessment reports, PESTEL news intelligence, and curriculum, module, handout, and quiz generation.
    - Designed a skill-based agent with progressive-disclosure middleware (templates loaded on demand via tool calls) and parallel section generation using LangGraph Send() with Tavily web search; map-reduce summarization for long news corpora.
    - Built parent–child (small-to-big) hybrid RAG on Milvus with Gemini embeddings, and a multi-format ingestion pipeline (PDF, DOCX, PPTX, XLSX) with Gemini multimodal OCR for scanned PDFs.
    - Added admin-managed, versioned prompt templates in PostgreSQL and smoke tests enforcing architectural boundaries.
  - Voice AI — SmartView Speech-to-Speech & Avatar Voice Assistant
    - Built an Indonesian voice assistant pipeline: Whisper speech-to-text → Elarch RAG → text-to-speech (VITS or cloned voice), letting users ask questions about archive documents by voice.
    - Reduced perceived latency with sentence-level streaming: LLM tokens are split into sentences (Indonesian-aware chunker) and synthesized in parallel, so audio starts playing before the full answer is generated.
    - Extended the Elarch RAG service with WebSocket voice chat and talking-avatar endpoints (speech-to-text, per-turn answer, and text-to-speech) for an interactive avatar experience.

- HCC K3 — AI CCTV Safety Compliance Platform (Computer Vision & MLOps)
  Built an end-to-end workplace-safety platform that detects missing personal protective equipment (PPE) on live CCTV, verifies detections with a vision-language model, alerts safety officers, and continuously retrains from human feedback.
  - Detector
    - Ultralytics YOLO on RTSP streams with automatic TensorRT FP16 export (engine cached in MinIO for reuse across replicas), body-zone PPE-to-person association, a custom IoU tracker with event grouping, and Redis lease-based camera sharding for horizontal scaling.
  - Reviewer
    - VLM-based verification (OpenAI-compatible endpoint via Ollama/vLLM) returning structured Pydantic verdicts to filter false positives.
  - Trainer
    - Queue-driven retraining pipeline (collect → train → evaluate → register), auto-triggered after 100 human corrections, with mAP50 gating, human promotion, and a versioned model registry in MinIO; detectors hot-reload promoted models.
  - API & Data
    - FastAPI control plane for cameras, areas of interest, detections, corrections, training, and compliance analytics; TimescaleDB hypertables with retention and compression, Alembic migrations, and a shared versioned library across services; containerized with Docker.

- Icon Cloud — JupyterHub Multi-User GPU Platform
  Deployed a production-grade multi-user notebook platform with isolated GPU allocation across two bare-metal servers, serving tiered GPU access via NVIDIA MIG on H200.
  - GPU Architecture
    - NVIDIA H200 141GB across 2 servers (8× GPUs each), with tiered MIG profiles — 3g.71gb (Premium), 2g.35gb (Standard), 1g.18gb (Basic) — giving each user a fully isolated GPU slice.
    - MIG instance UUIDs and availability tracked in PostgreSQL via an automated sync script.
  - Platform
    - JupyterHub with DockerSpawner injecting each user's MIG UUID via NVIDIA Container Runtime.
    - Custom CUDA 12.8 notebook images with group-based MIG assignment via dynamic JupyterHub configuration.
    - Traefik reverse proxy routing multi-server deployments under a single domain.
  - Backend Wrapper API
    - Built a FastAPI service for automated user lifecycle management across multi-server deployments.

PT Intelix Global Crossing **Jakarta, Indonesia**
AI & Data *June 2024 – March 2025*

- Chatbot Development
  - Cloud-based: Built a customer service chatbot using OpenAI as the base model, enhanced with LangChain for better adaptability, and deployed via a Flask API to ensure seamless integration and scalability.
  - On-premise: Engineered an advanced RAG-based chatbot leveraging LangChain, BAAI embeddings, ChromaDB, and Mistral-7B for natural conversations. Developed a JSON-based conversation history system and deployed it on an NVIDIA T4 GPU. Created multiple specialized bots, including a Customer Service Bot, SQL Agent, and Summarization Bot.
- Agent Assistant
  - Designed and deployed an LLM-powered assistant for live agents, providing real-time response suggestions to improve agent efficiency and consistency in customer interactions.
- AI QA/QM Assistant
  - Developed a QA/QM automation tool using OpenAI's LLM to streamline the quality assurance process: transcribed audio conversations to text, then summarized, analyzed sentiment, and auto-classified conversations using LLM pipelines.
- AI Decision Engine
  - Implemented machine learning models to enhance product intelligence, including customer segmentation, demand forecasting, and multi-label ticket classification.

PLN Icon Plus **Jakarta, Indonesia**
Data Scientist Intern *February 2024 – June 2024*

- **Anomaly Detection:** Detected anomalies in 10 years of customer data by clustering kWh usage and power with K-Means, selecting the number of clusters using the elbow method and silhouette score. Applied Median Absolute Deviation (MAD) within each cluster to set upper, middle, and lower bounds for anomaly identification.
  Technology / Tools: Python, Pandas, NumPy, Scikit-learn, SciPy, Seaborn, Matplotlib, Tableau.
- **Tsunami Modeling:** Researched tsunami wave modeling using COMCOT (Cornell Multi-grid Coupled Tsunami Model) to simulate the amplitude of earthquake-generated waves.
  Technology / Tools: gfortran, MATLAB, Python, Pandas, NumPy, Scikit-learn, Matplotlib.

AXA Mandiri Syariah **Bandung, Indonesia**
Financial Advisor *June 2022 – October 2023*

- Handled both regular and priority customers of Bank Syariah Indonesia (BSI) to assess their financial situations and needs.
- Analyzed customer needs and generated financial recommendations based on initial assessments.
- Provided after-sales service to BSI customers while effectively addressing customer complaints.
- Compiled achievement data of Financial Advisors across BSI branches in the Bandung Raya area and reported it daily, weekly, and monthly.

Lembaga Ilmu Pengetahuan Indonesia (LIPI) **Bandung, Indonesia**
Research Intern *May 2018 – July 2018*

- Conducted research on Visible Light Communication (VLC), studying wireless data transmission through visible light using light-emitting diodes (LEDs).
- Focused on improving data rates, expanding coverage range, enhancing reliability, and inferring nearby objects, and developed a miniature VLC model using Arduino UNO.

SKILLS

General Skills: Exploratory Data Analysis, Time Series Analysis, Machine Learning, Deep Learning, Large Language Models (LLM), Generative AI, Agentic AI, Retrieval-Augmented Generation (RAG), Computer Vision, Speech AI.
Programming Languages: Python, SQL, MATLAB.
LLM & Agent Frameworks: LangChain, LangGraph, Hugging Face Transformers, Ollama.
LLM Serving & Observability: vLLM, LiteLLM, Langfuse.
Models: Qwen3, Gemma, Gemini, OpenAI GPT, Llama, Mistral, Whisper, MMS-TTS, YOLO, BERT, VGG-16, ResNet50.
ML / DL Libraries: PyTorch, TensorFlow, Keras, Ultralytics, TensorRT, OpenCV, Scikit-learn, Pandas, NumPy, SciPy, Feature-engine, Matplotlib, Seaborn.

Backend & Apps: FastAPI, Flask, Streamlit, SQLAlchemy, Alembic, WebSocket, Server-Sent Events (SSE).
Databases, Search & Storage: PostgreSQL, TimescaleDB, SQL Server, Milvus, ChromaDB, Elasticsearch, Redis, MinIO.
Infrastructure & MLOps: Docker, Docker Compose, NVIDIA MIG, CUDA, JupyterHub, Traefik, Apache Airflow.
Visualization Tools: Tableau, Kibana.
Modeling Algorithms: Regression, Random Forest, Decision Trees, Neural Networks, Clustering, Dimensionality Reduction, NLP, Computer Vision, Time Series Analysis.
Others: Amazon Web Services (AWS Glue), Google BigQuery, Postman.

CERTIFICATIONS

Hacktiv8
Full Time Data Science — *Issued on February 2024*
Certificate

Boston Consulting Group
Data Science & Analytics Virtual Experience Program — *Issued on February 2024*
Certificate — *No expiration date*

HackerRank
Python (Basic) — *Issued on February 2024*
Certificate — *No expiration date*

HackerRank
SQL (Basic) — *Issued on February 2024*
Certificate — *No expiration date*

Digitalent Kominfo
Intro to Full Stack Developer — *Issued on August 2021*
Certificate — *No expiration date*